

Q:1 What is the prime objective of Software engineering

The prime objective of software engineering is to develop high quality software that is:

- reliable: work correctly & consistently.
- maintainable: easy to modify & enhance
- cost effective: within time & budget
- meets the user's requirements & expectations

The main goal is to deliver useful, reliable & maintainable software products in a systematic & disciplined way.

Q:2 Define software engineering paradigm.

A software engineering paradigm is a framework of principles, practice, methods and tools that provides a systematic and disciplined approach to software development from requirements to maintain.

It combines proven technique and methodologies, based on past experience and scientific principles, to manage the complexity of software development and to improve productivity, quality and maintainability of software.

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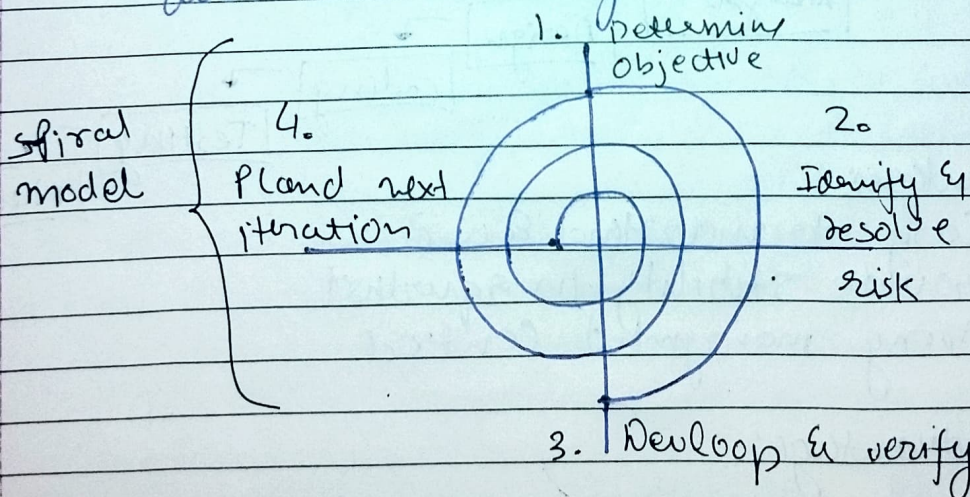
Q:3 what do you mean by spiral model?

Spiral model is a process model of software development which combines the iterative nature of iterative model and the systematic aspects of waterfall model.

Key points

- It is risk driven model
- Development is done in a series of iterations (spirals)
- Each iteration consists of four phases
 - Determine objective
 - Identify and resolve risk
 - Develop and verify
 - Plan the next iteration

→ Suitable for large and complex projects where risk are high



It is flexible model which allows changes and manages risks effectively

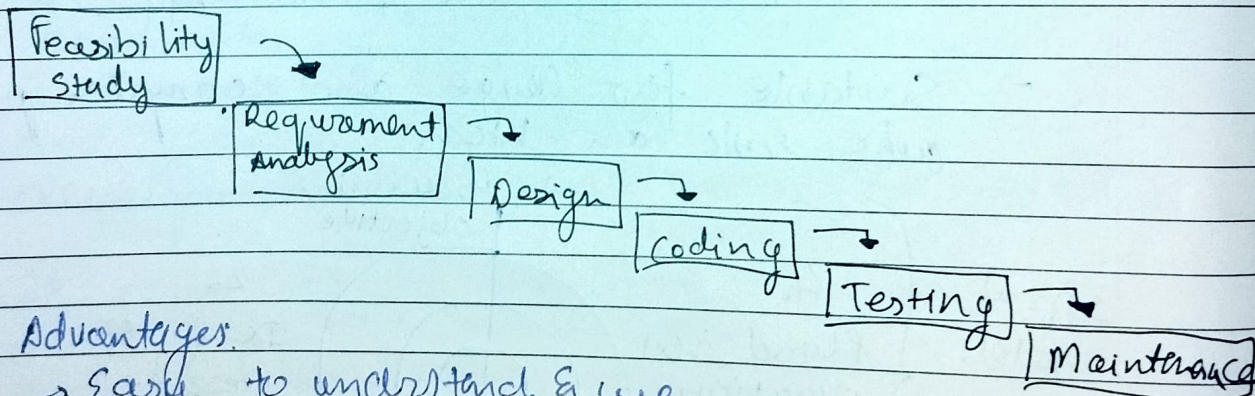
Q.4 Write a brief note on waterfall model.

Waterfall model is a linear and sequential approach to software. In this model each phase is completed before the next phase begins. It is also called a classical life cycle model.

Key points.

- it follows a fixed sequence of phase
- it is simple to understand & easy to manage
- it works well when requirements are clear and stable

Phases:



Advantages:

- Easy to understand & use
- provide stability in requirement
- strong management control

Disadvantage:

- Not stable for changing req.
- Integration is done in end.
- Little customer involvement.

Q5 Distinguish between process and method.

PROCESS	METHOD
→ Process is a set of activities performed in a specific order to achieve goal.	→ Methods are specific techniques, tools or procedure used to perform the activities in process.
→ It is a broader view (macro level).	→ It is a detailed view (micro level).
→ Defines overall workflow and phases of development.	→ Defines how to do the work.
→ Waterfall, Agile, Spiral.	→ UML, DFD, Testing.

Q6 Give the importance of software engineering.

- Helps in building high quality and reliable software.
- Improves productivity and reduces development time and cost.
- Provide proper structure, planning and documentation.
- Helps in managing complexity of large software.
- System Risk and improves.
- Security. Support teamwork and better communication.
- Leads to customer satisfaction by delivering software that meets real needs.

Q.7 Define software process. State the important feature of it.

Software process: State the important feature of a process.

Software process: It is a set of related activities, methods, practices and work product that are used to develop, operate & maintain. It provides a structured way to manage the development and ensure that software is built with quality, within time & cost constraints.

Important features:

- ordered set of activities:
 - Define the sequence of task to be performed
 - Define input & output.
 - Specify what is required and what is provided.
- Roles and responsibilities.
 - defines who does what
- Milestone and deliverables
 - Helps in tracking progress and completion
- Repeatable and measurable
 - can be followed consistently and its progress can be measured
- adaptable

Q.8 Write any two characteristics of software as a product.

→ Software is Engineered / Developed not manufactured
 Unlike hardware products, where raw material are assembled & physically manufactured cost dominates, software is intellectual & creative activity

Portability: can be moved from one platform to another

Q.9 List the process maturity level in SEI's CMM.

SEI's comparability maturity model (CMM):

Initial	level - I
Repeatable	level - II
Defined	level - III
Managed	level - IV
Optimizing	level - V

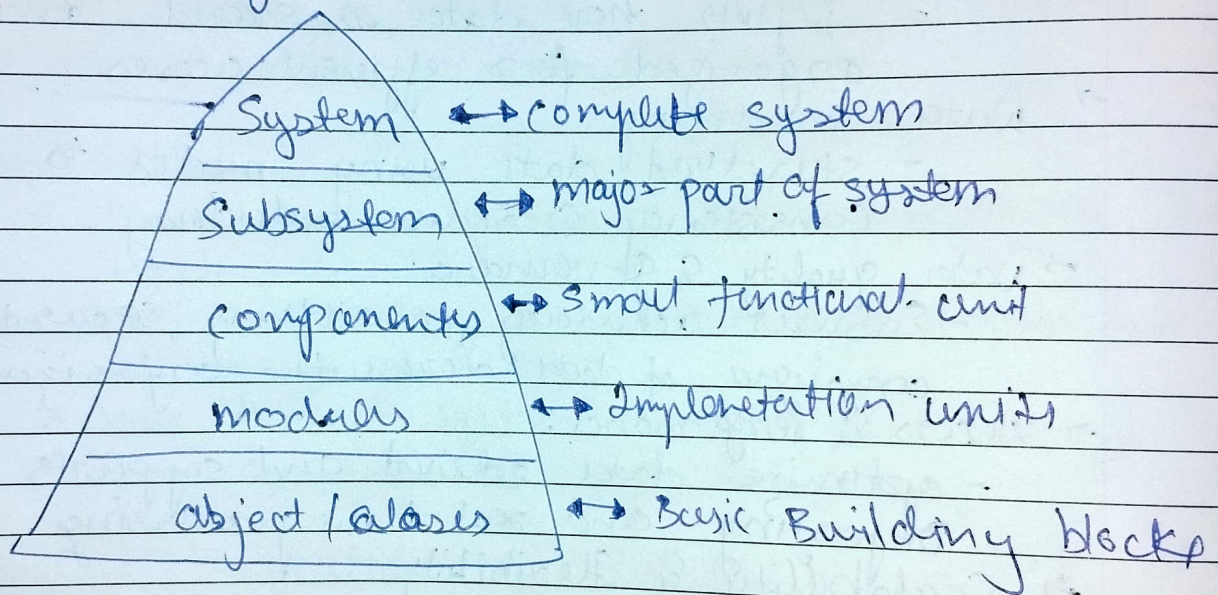
Q.12 Define System modeling.

System modeling is a process of creating a simplified representation of a real-world system to understand, analyze and design its structure, behavior and components before actual implementation.

Q.13 State System Engineering Hierarchy.

The system Engineering Hierarchy is a structured approach that shows the relationship between different levels of abstraction in a system, from the highest level to lowest level.

Hierarchy



~~This~~ ⇒ Ensures proper decomposition, modularity and ease of management in system development

Q.11 What are the functions of data architecture?

Functions of data architecture:

- Data integration:
 - combines data from multiple source into a unified view
- Data storage
 - defines how data is stored and organized for efficient access
- Data modeling
 - structured data using model to ensure consistency & reduce redundancy
- Data quality & Governance:
 - Ensures accuracy, consistency, security & compliance of data across the organization
- Access & Performance:
 - optimize data retrieval and supports analytics reporting and real-time processing
- Scalability & Flexibility
 - Handles growing data volume & adapts to change business need and technologies

Q.14 Mention some of factor to be consider during System modeling.

- Define System boundary
- Identify inputs and outputs
- Identify the major components / subsystem
- specify the data and control flow
- consider performance, reliability & security req.
- model the behavior of system
- Validate the model with user requirement.

Q: 10 Distinguish between verification & validation

Verification

- ↳ it answers the question "Are we building the product right?"
- Involves checking and reviewing the product against specification, standards and requirements
- Done through activities like review, testing
- Mostly static - no execution

Validation

- ↳ it answers the question: "Are we building the right product?"
- Involves checking the product against user need and intended use
- Done through system testing, user testing
- Dynamic, - execution of code

Q: 15 Define Verification & Validation

Verification: → process of checking whether product
 1) built according to req & specs.
 → it swears around
 1) product built right

Validation: process of checking whether product
 1) actually what user needs.
 → is product right built?